



Examen Parcial – 06/5/16

Apellido y nombre:

DNI:

1.

- a. Usando notación indicial, mostrar que, para vectores arbitrarios $\mathbf{a}$, $\mathbf{b}$, se verifica:

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})^2 = (ab)^2$$

donde a, b son los módulos de $\mathbf{a}$, $\mathbf{b}$ respectivamente.

- b. Mostrar que $e_{ijk}\sigma_{jk} = 0$, donde e_{ijk} es el símbolo de permutación y $\sigma_{jk} = \sigma_{kj}$ es un tensor simétrico.

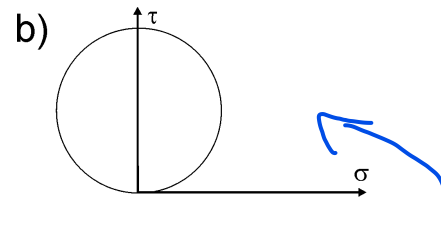
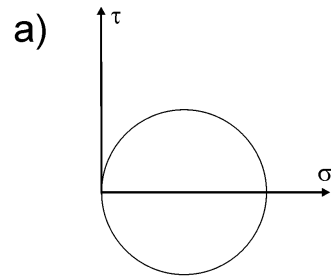
2. Para la función $A = A_{ij}x_i x_j$, calcular usando notación indicial cuanto vale $\frac{\partial A}{\partial x_i}$ y $\frac{\partial^2 A}{\partial x_i \partial x_j}$, donde los coeficientes A_{ij} son constantes.

3. Un campo de tensiones en una base Cartesiana está dado por la matriz:

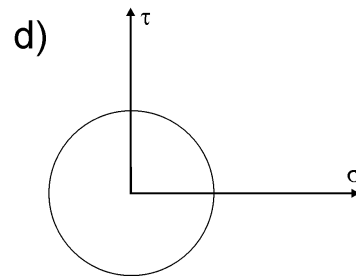
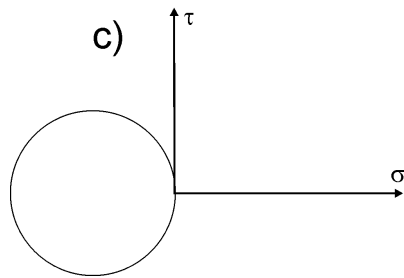
$$T = \begin{bmatrix} (1-x_1^2)x_2 + \frac{2}{3}x_2^3 & -(4-x_2^2)x_1 & 0 \\ -(4-x_2^2)x_1 & -\frac{1}{3}(x_2^3 - 12x_2) & 0 \\ 0 & 0 & (3-x_1^2)x_2 \end{bmatrix}$$

Mostrar que las ecuaciones de equilibrio se satisfacen en todo punto con fuerzas de volumen nulas.

4. Para el tensor de tensiones planteado en el ejercicio anterior, evaluado en el punto $P(2, -1, 6)$:
- Determinar el vector de tensiones en el plano cuya ecuación es $3x_1 + 6x_2 + 2x_3 = 12$.
 - Determinar las tensiones principales.
5. En cada una de las situaciones siguientes, indique si el círculo de Mohr dibujado tiene realidad física, y cuando sea así, dé el valor del corte máximo en función del valor de las tensiones principales representadas. En todos los casos, justifique en forma clara y precisa su respuesta.



DEBE ESTAR EL CENTRO SOBRE EL EJE SIGMA O



①

$$1) a) (\underline{a} \times \underline{b}) \cdot (\underline{a} \times \underline{b}) + (\underline{a} \cdot \underline{b})^2 = (ab)^2$$

$$\begin{aligned} \varepsilon_{ijk} a_j b_k \varepsilon_{ilm} a_l b_m + a_i b_i a_j b_j &= \\ = \delta_{jl} \delta_{km} a_j b_k a_l b_m - \delta_{jm} \delta_{kl} a_j b_k a_l b_m + & \\ + a_i b_i a_j b_j &= \\ = a_l b_m a_l b_m - \cancel{a_m b_l a_l b_m} + \cancel{a_i b_i a_j b_j} &= \\ = a_l a_l b_m b_m = a^2 b^2 = (ab)^2 & \end{aligned}$$

$$b) \varepsilon_{ijk} \sigma_{jk} = 0$$

$$\sigma_{jk} = \sigma_{kj} \implies \sigma_{jk} = \frac{\sigma_{jk} + \sigma_{kj}}{2}$$

$$\begin{aligned} \varepsilon_{ijk} \sigma_{jk} &= \varepsilon_{ijk} \frac{\sigma_{jk}}{2} + \varepsilon_{ijk} \frac{\sigma_{kj}}{2} = \\ &= \varepsilon_{ijk} \frac{\sigma_{jk}}{2} + \varepsilon_{ikj} \frac{\sigma_{jk}}{2} = \\ &= \varepsilon_{ijk} \frac{\sigma_{jk}}{2} - \varepsilon_{ijk} \frac{\sigma_{jk}}{2} = 0 \end{aligned}$$

(2)

$$2) \quad A = A_{ij} x_i x_j = A_{lm} x_l x_m$$

$$\frac{\partial A}{\partial x_i} = \frac{\partial (A_{lm} x_l x_m)}{\partial x_i} = A_{lm} \frac{\partial x_l}{\partial x_i} x_m +$$

$$+ A_{lm} x_l \frac{\partial x_m}{\partial x_i} =$$

$$= A_{lm} \delta_{li} x_m + A_{lm} x_l \delta_{mi} =$$

$$= A_{im} x_m + A_{li} x_l$$

$$\frac{\partial^2 A}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{\partial A}{\partial x_i} \right) = \frac{\partial}{\partial x_j} (A_{im} x_m + A_{li} x_l) =$$

$$= A_{im} \frac{\partial x_m}{\partial x_j} + A_{li} \frac{\partial x_l}{\partial x_j} =$$

$$= A_{im} \delta_{mj} + A_{li} \delta_{lj} = A_{ij} + A_{ji}$$

(3)

$$3) \quad T = \begin{bmatrix} (1-x_1^2)x_2 + \frac{2}{3}x_2^3 & -(1-x_2^2)x_1 & 0 \\ -(1-x_2^2)x_1 & -\frac{1}{3}(x_2^3 - 12x_2) & 0 \\ 0 & 0 & (3-x_1^2)x_2 \end{bmatrix}$$

$$T_{ij,j} = ?$$

$$\frac{\partial T_{11}}{\partial x_1} + \frac{\partial T_{12}}{\partial x_2} + \frac{\partial T_{13}}{\partial x_3} = -2x_1x_2 + 2x_1x_2 = 0$$

$$\frac{\partial T_{21}}{\partial x_1} + \frac{\partial T_{22}}{\partial x_2} + \frac{\partial T_{23}}{\partial x_3} = -(1-x_2^2) - x_2^2 + 1 = 0$$

$$\frac{\partial T_{31}}{\partial x_1} + \frac{\partial T_{32}}{\partial x_2} + \frac{\partial T_{33}}{\partial x_3} = 0$$

1

$$1) P = (2, -1, 6)$$

$$\underline{T} = \begin{pmatrix} (1-4)(-1) - \frac{2}{3} & -(4-1)2 & 0 \\ -(4-1)2 & -\frac{1}{3}(-1+12) & 0 \\ 0 & 0 & (-1)(-1) \end{pmatrix} =$$

$$\underline{T} = \begin{pmatrix} 7/3 & -6 & 0 \\ -6 & -11/3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2) 3x_1 + 6x_2 + 2x_3 = 12$$

Vert perp plane:

$$\underline{u} = \begin{pmatrix} 3 \\ 6 \\ 2 \end{pmatrix}$$

$$\underline{n} = \frac{\underline{u}}{\|\underline{u}\|} = \begin{pmatrix} 3/7 \\ 6/7 \\ 2/7 \end{pmatrix}$$

$$\|\underline{u}\| = \sqrt{9+36+4} = 7$$

$$\underline{t} = \underline{T} \underline{n} = \begin{pmatrix} 7/3 & -6 & 0 \\ -6 & -11/3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3/7 \\ 6/7 \\ 2/7 \end{pmatrix} = \begin{pmatrix} -29/7 \\ -18\frac{1}{7} - \frac{22}{7} \\ 2/7 \end{pmatrix}$$

$$\underline{t} = \begin{pmatrix} -29/7 \\ -40/7 \\ 2/7 \end{pmatrix}$$

$$b) \det \begin{pmatrix} (7/3 - \lambda) & -6 & 0 \\ -6 & (-11/3 - \lambda) & 0 \\ 0 & 0 & (1 - \lambda) \end{pmatrix} = 0 \Rightarrow \lambda_1 = 1$$

$$(7/3 - \lambda)(-11/3 - \lambda) - 36 = 0$$

$$-\frac{77}{9} - \frac{7}{3}\lambda + \frac{11}{3}\lambda + \lambda^2 - \frac{324}{9} = 0$$

$$\lambda^2 + \frac{4}{3}\lambda - \frac{401}{9} = 0$$

$$\lambda_{2,3} = \frac{-\frac{4}{3} \pm \sqrt{\frac{16}{9} + \frac{1604}{9}}}{2} = \frac{-\frac{4}{3} \pm \frac{18}{3}\sqrt{5}}{2} =$$

$$= -\frac{2}{3} \pm 6\sqrt{5}$$

$$\therefore \lambda_1 = 1$$

$$\lambda_2 = -\frac{2}{3} + 3\sqrt{5} = 6,03153\dots$$

$$\lambda_3 = -\frac{2}{3} - 3\sqrt{5} = -7,37187\dots$$

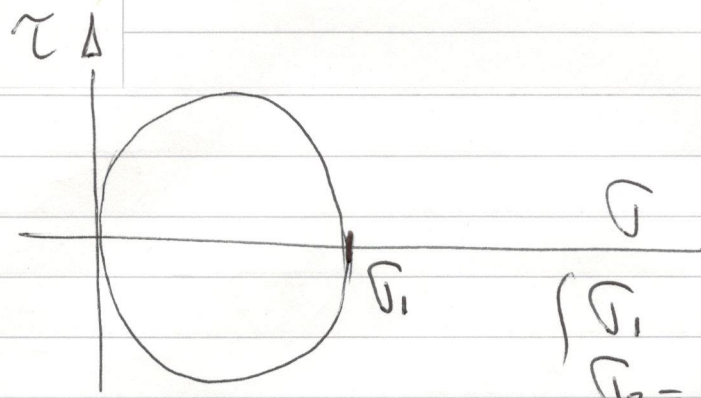
Ordered, de mayor a menor:

$$\lambda_1 = -\frac{2}{3} + 3\sqrt{5}$$

$$\lambda_2 = 1$$

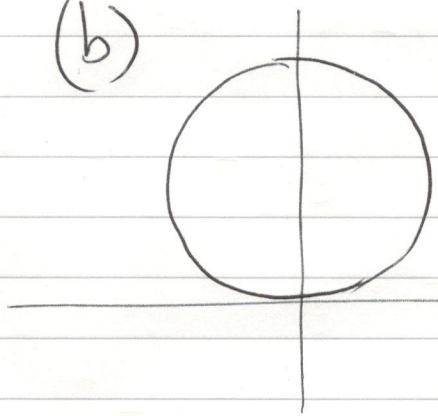
$$\lambda_3 = -\frac{2}{3} - 3\sqrt{5}$$

5)(a)



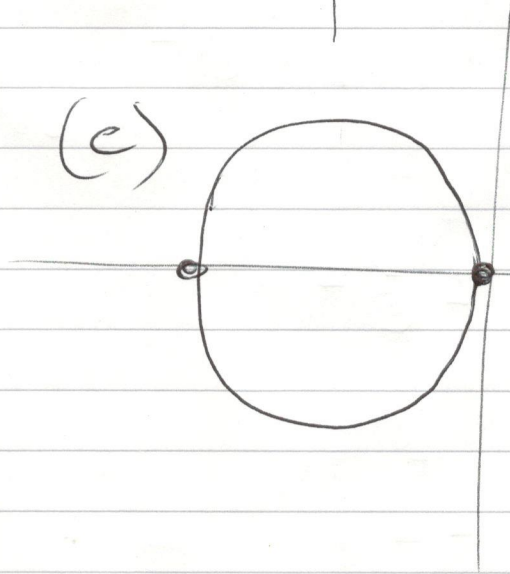
$$\begin{aligned} \sigma_1 \\ \sigma_2 = 0 \\ \tau_{max} = \sigma_1 / 2 \end{aligned}$$

(b)



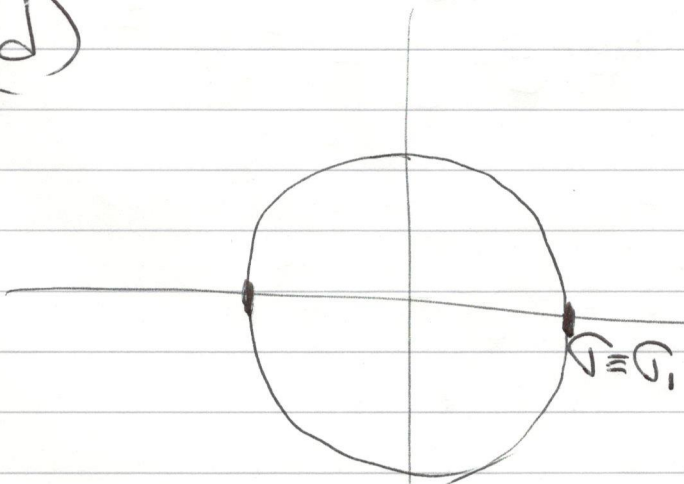
INCORRECT

(c)



$$\begin{aligned} \sigma_1 &= 0 \\ \sigma_2 &= \sigma_2 \\ \tau_{max} &= -\frac{\sigma_2}{2} \end{aligned}$$

(d)



$$\begin{aligned} \sigma_1 &= \sigma \\ \sigma_2 &= -\sigma \\ \tau_{max} &= \sigma = \sigma_1 \end{aligned}$$